

Medical Sensors
Project – Literature Review

Review on Traditional and Deep Learning Methods for Diabetic Retinopathy detection and classification

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Diabetic Retinopathy

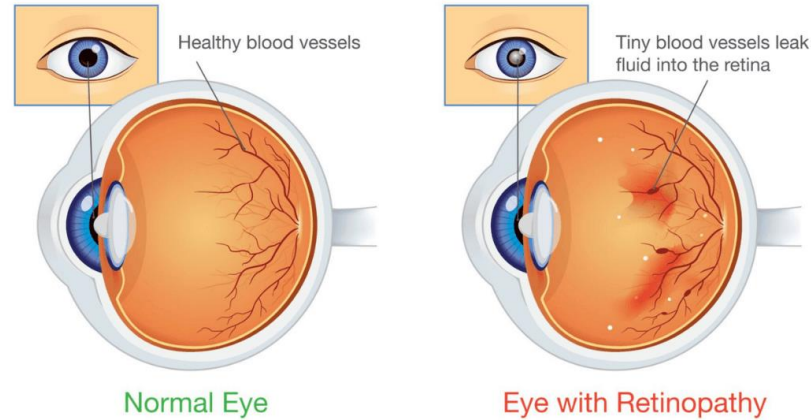


Figure 1: Anatomy of Diabetic Retinopathy [1]

- Diabetic Retinopathy (DR) is a complication of diabetes that causes the blood vessels of the retina to swell and to leak fluids and blood.
- The eye has many tiny blood vessels in its retina, that senses light and sends signals to the brain.
- High levels of sugar in the blood, as seen in diabetes, can make blood vessels in eye weak and damaged over time.
- Early detection and treatment are necessary to delay or avoid vision deterioration and vision loss.
- Computer-aided diagnosis system has achieved better performance in medical image analysis and classification compared to manual diagnosis methods

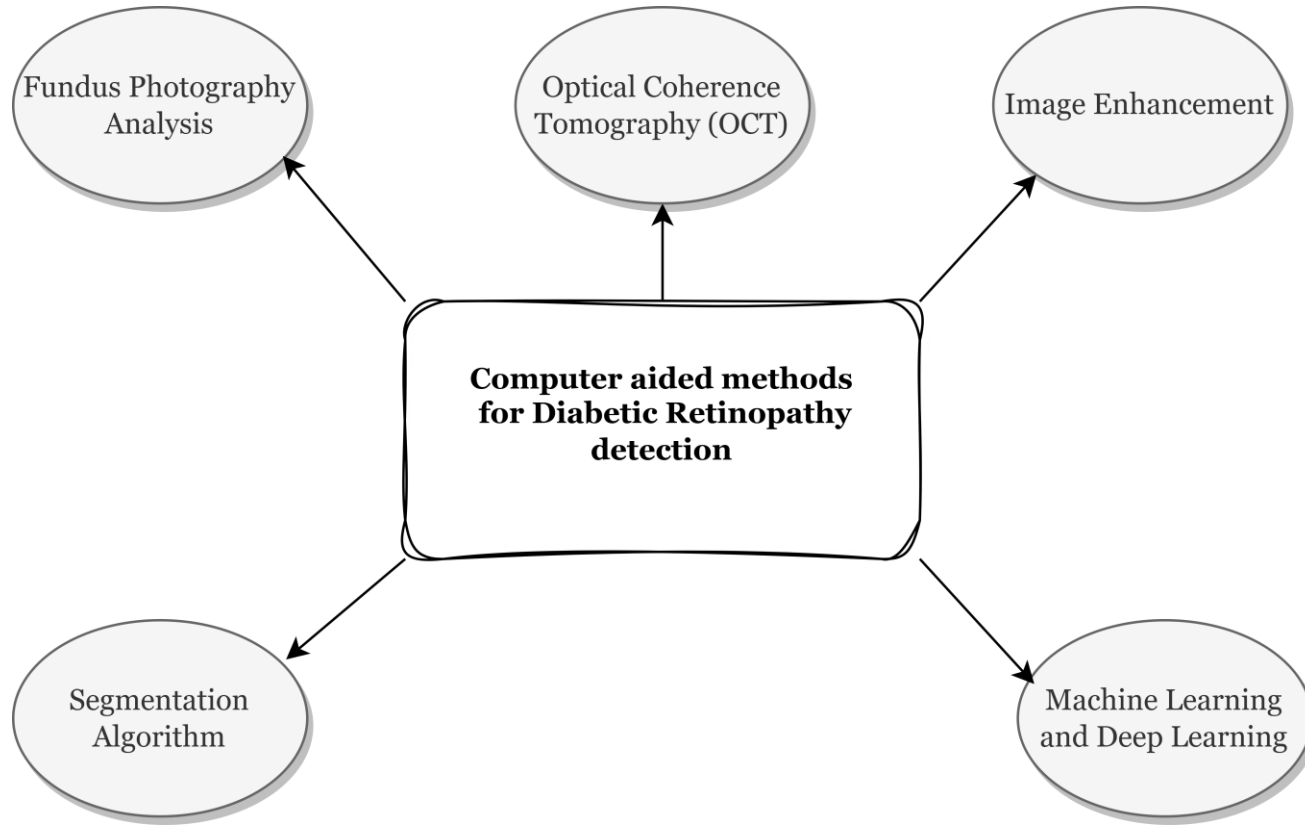


Figure 2: Computer aided methods for DR detection

Table 1: Commonly used datasets for Diabetic Retinopathy detection [2]

Datasets	No. of images	Training Set	Test Set	Image Size
DiaretDB1	89	28 images	61 images	1500 x 1152 pixel
Kaggle	88,702	35,126 images	53,576 images	Different image resolution
DRIVE	40	20 images	20 images	565 x 584 pixel
DDR	13,673	6835 images	4105 images	Different image resolution
IDRID	516	413 images	103 images	4288 x 2848 pixels
ROC	100	50 images	50 images	Different image resolution

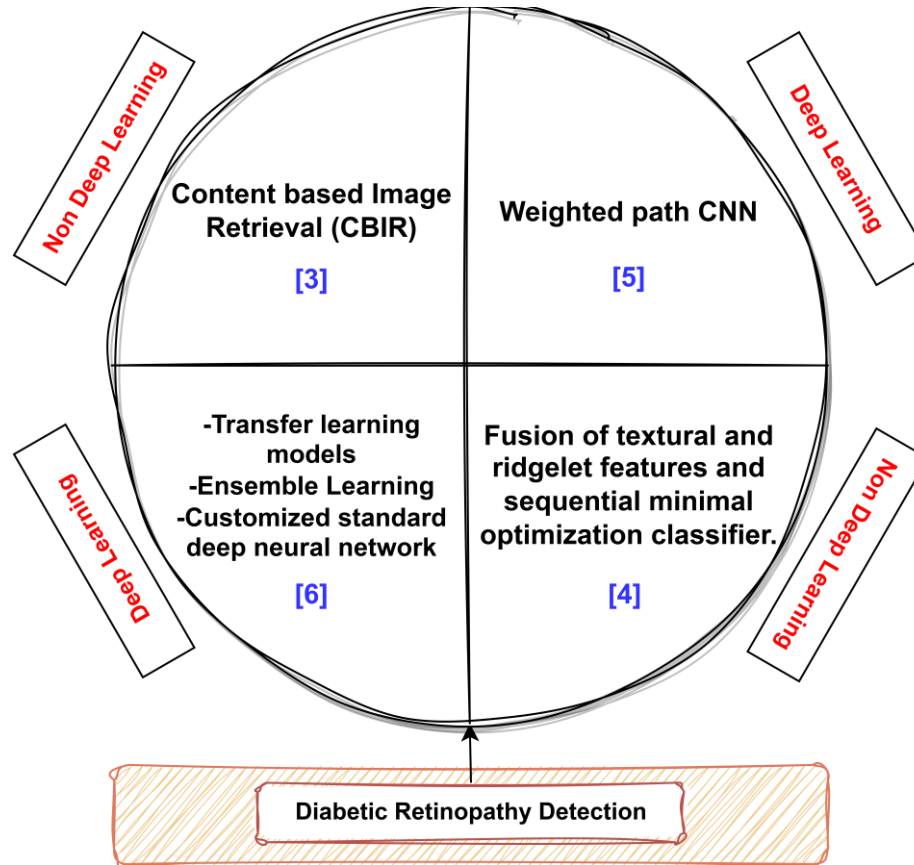


Figure 3: Literature Review of 4 methods for DR detection

Non-Deep Learning: Content Based Image Retrieval (CBIR)

Chaum E et. al, Retina [3]

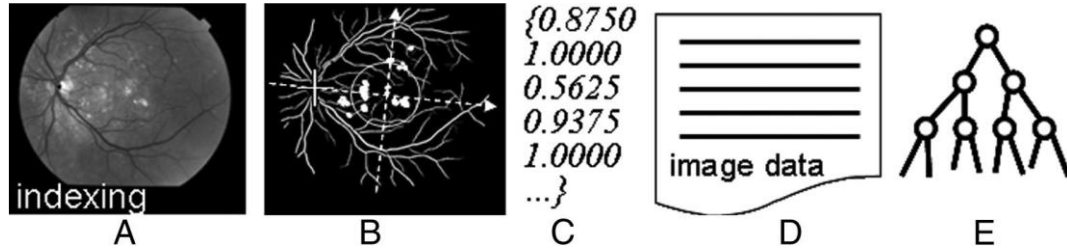


Figure 4: Indexing process

CBIR process

- A) Sample Image is taken for analysis.
- B) The image is analyzed to locate anatomic structures and pathologic lesions.
- C) Statistical features are extracted from these structures. These statistical features comprise a number matrix that describe shapes, texture, orientation, etc., of the structures and lesions that have been found.
- D) A table of these features is built (i.e., a database library) representing the image population.
- E) An indexing tree is generated that facilitates rapid searching.

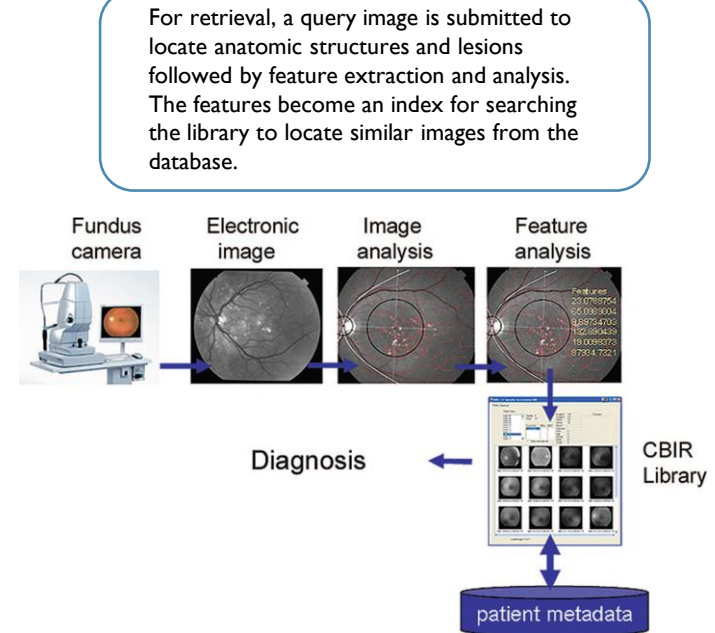


Figure 5: Retrieval process

Non-Deep Learning: Content Based Image Retrieval (CBIR)

Chaum E et. al, Retina [3]

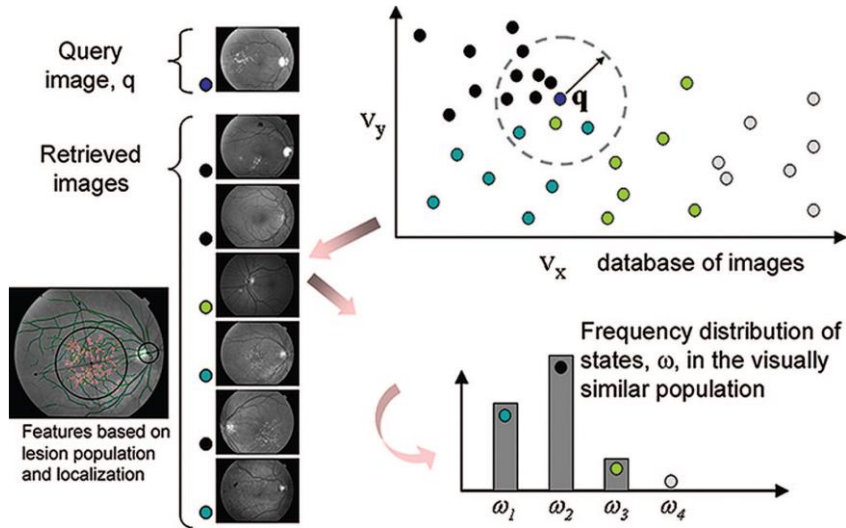


Figure 6: Retrieval and analysis process for a query image

	Norm	AMD	PDR	NPDR	Other	Unknown	Performance (%)
Norm	40	0	0	0	5	5	88.9
AMD	1	61	0	1	6	12	88.4
PDR	0	0	14	1	0	7	93.3
NPDR	0	0	0	46	5	7	90.2
Other	6	4	0	1	154	17	93.3
Purity (%)	85.1	93.9	100.0	93.9	90.6	12.2	91.3

Figure 6: Confusion Matrix Detailing Classification Performance Over Four Primary Disease Categories

Results

The diagnostic sensitivity for all stratified levels of age-related macular degeneration (AMD) ranged from 75% to 100%.

The sensitivity of detection and accuracy for proliferative diabetic retinopathy (PDR) ranged from 75% to 91.7%.

The sensitivity of detection and accuracy for nonproliferative diabetic retinopathy (NPDR) ranged from 75% to 94.7%.

The overall purity of the diagnosis (specificity) for all disease states in the dataset was 91.3%.

Non-Deep Learning: Fusion of textural and ridgelet features and sequential minimal optimization classifier.

Ramasamy LK et. al, PeerJ Comput Sci [4]

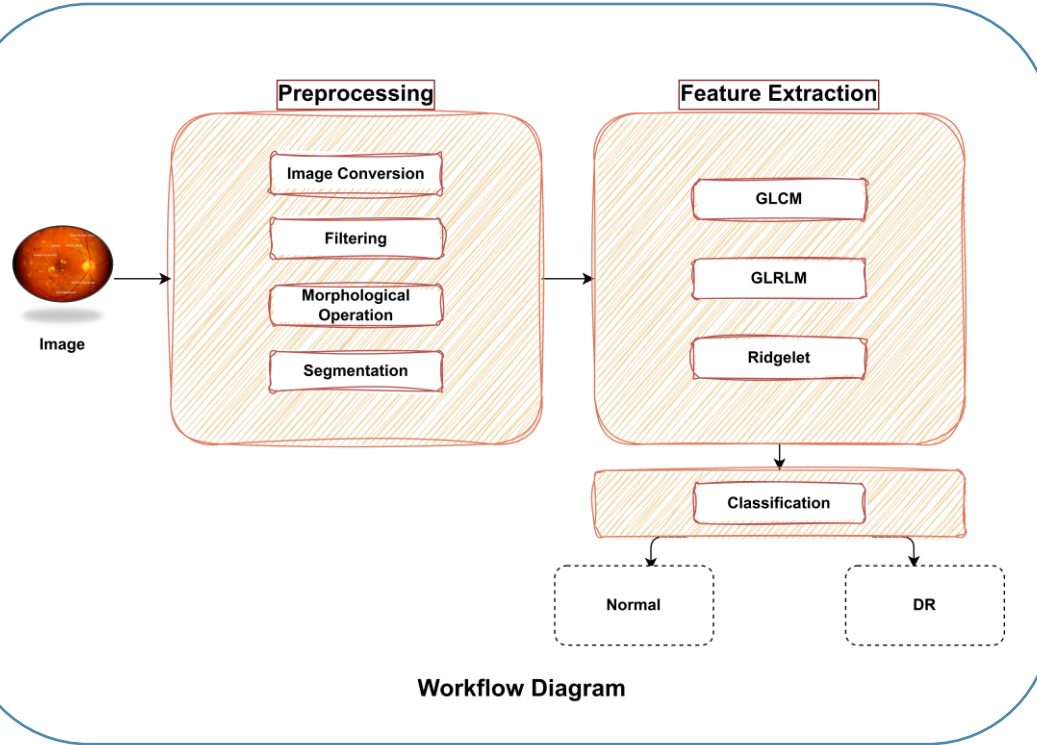


Figure 7: Proposed Algorithm

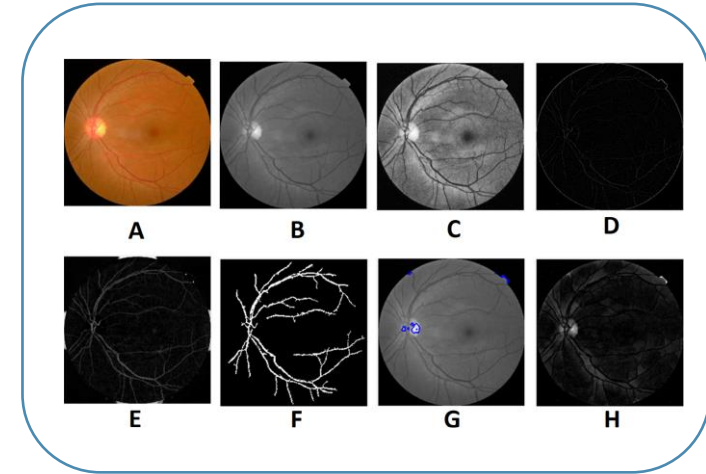


Figure 8: Preprocessed result of normal retina image:

- (A) input image
- (B) green channel
- (C) histogram enhanced
- (D) filtered image
- (E) after bottom hat transform
- (F) after top hat Transform
- (G) blood vessels segmented
- (H) contours enhanced.

Non-Deep Learning: Fusion of textural and ridgelet features and sequential minimal optimization classifier.

Feature Extraction

Gray Level Co-occurrence Matrix (GLCM): GLCM focuses on the spatial relationships between pixel intensities in an image. It measures how often pairs of specific gray levels occur together at certain distances and angles within an image.

Gray Level Run Length Matrix (GLRLM): GLRLM characterizes the lengths of consecutive runs of pixels with the same gray level. It is useful for capturing information about the size and continuity of specific features within an image, such as lines or textures.

Continuous ridgelet transform (CRT): CRT decomposes an image or signal into a set of ridge-like features, which are elongated and oriented in specific directions. These ridgelets are particularly effective in capturing information along these preferred directions.

Classification

Sequential Minimal Optimization (SMO)

SMO is a straightforward algorithm that uses only two Lagrange multipliers at each iteration to move the chunking process to the nearest possible expression.

It determines the optimal value for these multipliers and updates the Support Vector Machine (SVM) until it fixes the whole QP problem.

The benefit of SMO is that the optimization subproblem can be solved analytically with two Lagrange multipliers.

Ramasamy LK et. al, PeerJ Comput Sci [4]

Dataset	Feature Extraction Time (s)
DIARETDB1	131.56
KAGGLE	159.19

Dataset	SE (%)	SP (%)	ACC (%)	F-score
DIARETDB1	98.87	95.24	97.05	0.969
KAGGLE	90.9	91.0	91.0	0.909

Deep Learning: Referable diabetic retinopathy identification from eye fundus images with weighted path for convolutional neural network.

Yi-Peng Liu et. al, Ai in Medicine[5]

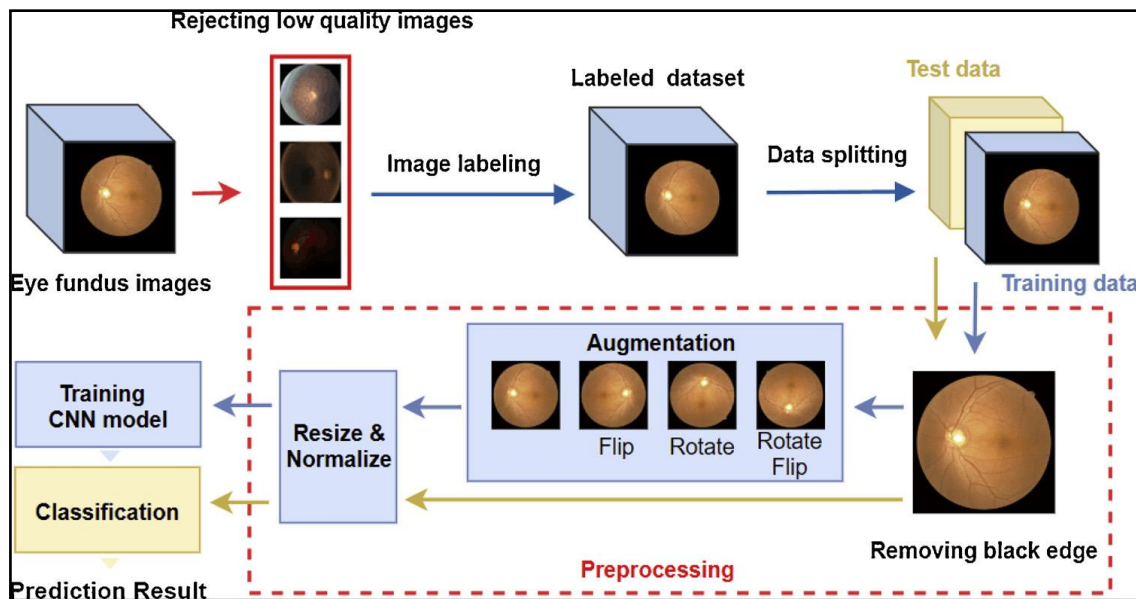


Figure 8: Workflow Diagram

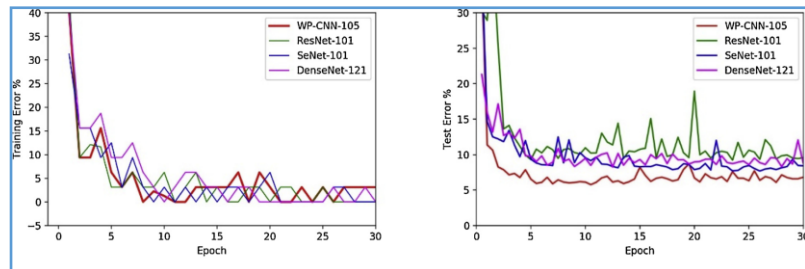
- As CNN models are made more deeper, the width of the network also increase and there is a greater probability of overfitting.
- Adding a weighted path in CNN can avoid model redundancy and strengthen the feature expression at the same time.
- Weighted Path (WP) assigns a weight coefficient to each path in the CNN network, which is evaluated randomly in the initialization process and updated during the network training process.
- The path value is then multiplied by its corresponding coefficient and the average of these weighted paths is then computed.

Weighted Path CNN

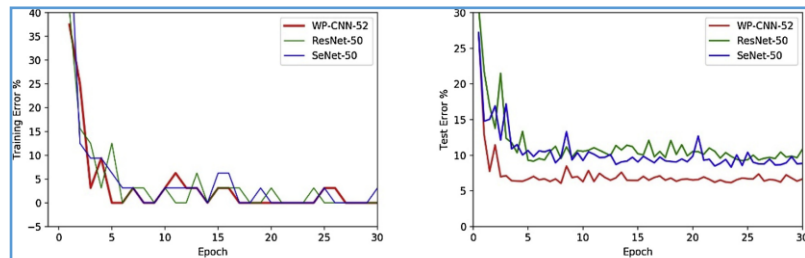
$$x_i = \frac{K_1 \times conv_1(x_{i-1}) + K_2 \times conv_2(x_{i-1}) + \dots + K_n \times conv_n(x_{i-1})}{n}$$

Deep Learning: Referable diabetic retinopathy identification from eye fundus images with weighted path for convolutional neural network.

Yi-Peng Liu et. al, Ai in Medicine[5]



The training and test process of different models with approximately 100 layers.



The training and test process of different models with approximately 50 layers.

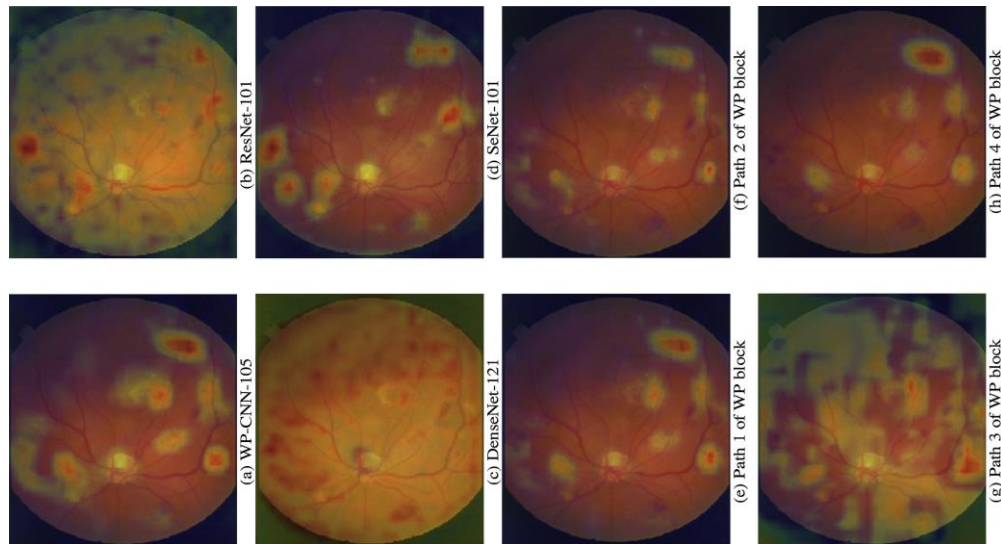


Figure 8: Fig: The feature map of the subgraph model overlapping with the test image. (a) WP-CNN-105 (b) ResNet-101 (c) DenseNet-121 (d) SeNet-101 (e) Path 1 of WP block (f) Path 2 of WP block (g) Path 3 of WP block (h) Path 4 of WP block.

Deep Learning: Referable diabetic retinopathy identification from eye fundus images with weighted path for convolutional neural network.

Yi-Peng Liu et. al, Ai in Medicine[5]

Results

Network	Accuracy	AUC	F1-score
ResNet-50	91.09%	0.9585	0.8606
ResNet-101	91.22%	0.9607	0.8594
DenseNet-121	92.39%	0.9676	0.8793
SeNet-50	92.21%	0.9619	0.8726
SeNet-101	92.55%	0.9703	0.8855
WP-CNN-32	93.80%	0.9797	0.9010
WP-CNN-52	94.04%	0.9819	0.9006
WP-CNN-105	94.23%	0.9823	0.9087

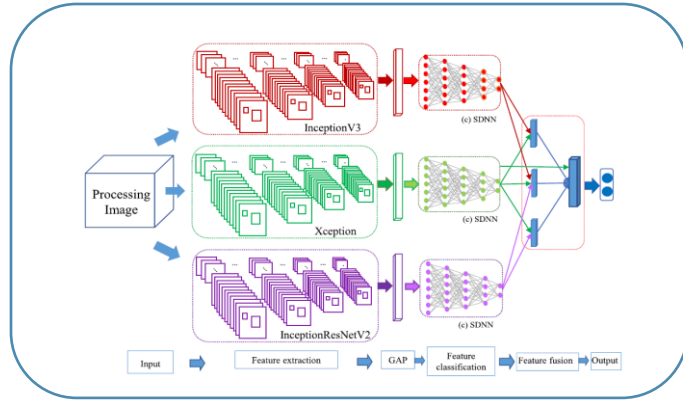
Networks performance on the referable diabetic retinopathy identification task.

Network	Accuracy	AUC	F1-score
ResNet-101	81.68%	0.878	0.855
DenseNet-121	85.50%	0.902	0.887
SeNet-101	83.21%	0.889	0.867
WP-CNN-105	90.84%	0.951	0.934

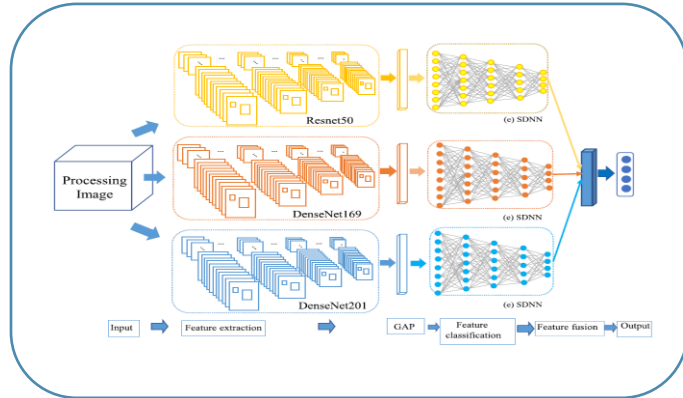
Performance of each model on STRAE dataset

Deep Learning: Automated identification and grading system of diabetic retinopathy using deep neural networks.

Wei Zhang et. al, Knowledge based systems[6]



Identification System



Grading System

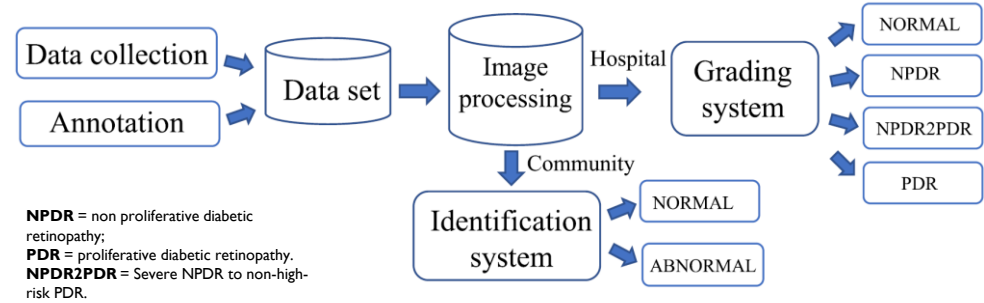


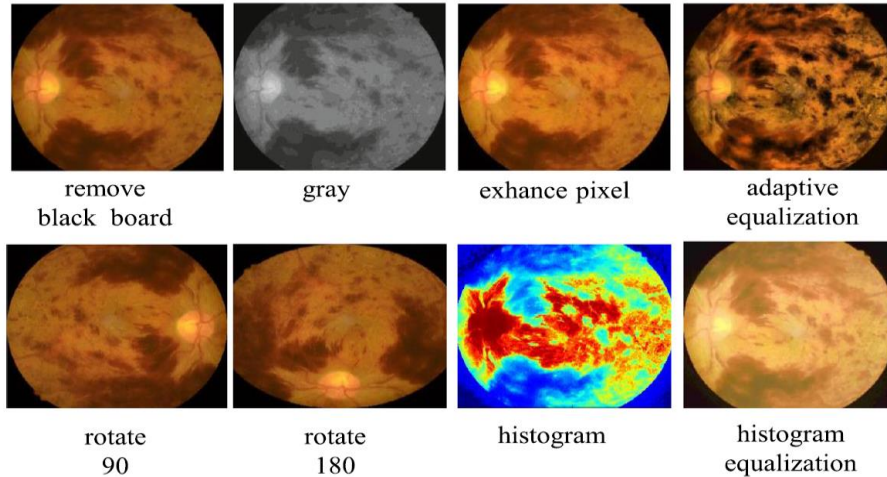
Figure 8: Overall Framework

Overall Framework

- Input Data
- Data pre-processing
- Single model feature extraction
- Single model feature classification
- Multi model feature fusion
- Output results

Deep Learning: Automated identification and grading system of diabetic retinopathy using deep neural networks.

Wei Zhang et. al, Knowledge based systems[6]



Data pre-processing

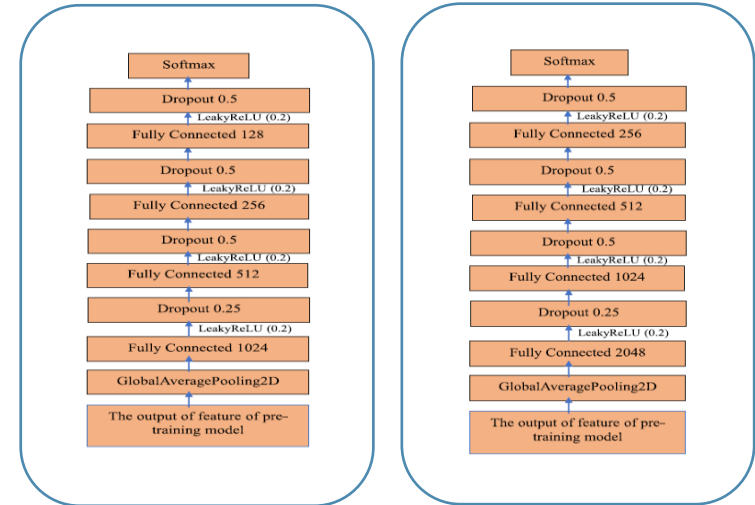


Figure 9: (A)SDNN for Identification system (B) SDNN for Grading system

- The first part of the neural network is a feature extractor comprising a pre-training model that was initialized via transfer learning which involved the removal of the top layer.
- The second part was a classifier that made predictions on the basis of the features of Diabetic Retinopathy; it was implemented by a customized stand deep neural network with training from scratch.
- Input to the customized standard deep neural network(SDNNs) was given by the output of the feature extractors.
- SDNNs have the same network depth but different parameters.

Deep Learning: Automated identification and grading system of diabetic retinopathy using deep neural networks.

Wei Zhang et. al, Knowledge based systems[6]

Single Model Feature Classification Results

Class distribution and classifications report of the identification system.

	Training	Validation	Testing	
Normal	1189	132	348	
Abnormal	1567	174	423	
Total	2756	306	771	
	Precision	Recall	F1_score	Support
Normal	0.97	0.98	0.97	348
Abnormal	0.98	0.97	0.98	423
ave/total	0.98	0.98	0.98	771

Identification System

Class distribution and classifications report of the grading system.

	Training	Validation	Testing	
Normal	1189	132	348	
npdr	469	52	114	
npdr2pdr	875	97	240	
pdr	224	25	69	
Total	2756	306	771	
	Precision	Recall	F1_score	Support
Normal	0.98	0.99	0.98	348
npdr	0.92	0.88	0.9	114
npdr2pdr	0.96	0.98	0.97	240
pdr	1	0.94	0.97	69
ave/total	0.96	0.96	0.96	771

Grading System

Deep Learning: Automated identification and grading system of diabetic retinopathy using deep neural networks.

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Multi Model Feature Fusion Results (Identification system)

Metrics of the identification system ensembles.

Model	Accuracy	Precision	Sensitivity	Specificity	Auc	Kappa	F1_score	F_{β_score}	Youden's index
Xception ^a	0.9521	0.9491	0.9321	0.9782	0.9551	0.9032	0.9523	0.9503	0.9101
Baseline:Resnet 50	0.9342	0.9333	0.9332	0.9351	0.9342	0.9231	0.9322	0.9223	0.9223
0:Xception ^b	0.9715	0.9711	0.9740	0.9683	0.9712	0.9424	0.9715	0.9712	0.9424
1:InceptionV3	0.9676	0.9667	0.9622	0.9741	0.9682	0.9676	0.9669	0.9346	0.9363
2:IncepresV2 ^c	0.9442	0.9435	0.9125	0.9828	0.9476	0.9444	0.9433	0.8883	0.8953
Ens(01) ^d	0.9767	0.9759	0.9716	0.9828	0.9772	0.9529	0.9767	0.9761	0.9544
Ens(02) ^e	0.9689	0.9677	0.9574	0.9828	0.9701	0.9373	0.9689	0.9680	0.9402
Ens(12) ^f	0.9598	0.9585	0.9456	0.9770	0.9613	0.9191	0.9599	0.9588	0.9226
3:Ens(012) ^g	0.9702	0.9691	0.9622	0.9799	0.9710	0.9340	0.9702	0.9694	0.9421
Ens(012,01) ^h	0.9689	0.9678	0.9598	0.9799	0.9698	0.9373	0.9689	0.9681	0.9397
Ens(012,02) ⁱ	0.9662	0.9652	0.9574	0.9770	0.9672	0.9321	0.9663	0.9655	0.9345
Ens(0,01,02) ^j	0.9753	0.9750	0.9764	0.9742	0.9752	0.9503	0.9753	0.9750	0.9505
Ens(0,2,01,02) ^k	0.9702	0.9691	0.9621	0.9799	0.9710	0.9399	0.9710	0.9694	0.9420
Ens(0,1,01,02) ^l	0.9740	0.9732	0.9693	0.9799	0.9746	0.9477	0.9741	0.9735	0.9492
Ens(0,01,02,3) ^m	0.9767	0.9760	0.9740	0.9799	0.9769	0.9529	0.9767	0.9762	0.9539
Ens(0,2,01,02,3) ⁿ	0.9701	0.9691	0.9622	0.9799	0.9710	0.9399	0.9702	0.9694	0.9421
Ens(0,1,01,02,3) ^o	0.9741	0.9732	0.9693	0.9799	0.9745	0.9477	0.9741	0.9735	0.9492
Ens (0,01,02,012) ^p	0.9767	0.9760	0.9764	0.9800	0.9862	0.9530	0.9769	0.9762	0.9540

^aEns(01) = Average of InceptionV3 and IncepresV2; Ens(01)_m = max of them.

^bEns(02) = Average of InceptionV3 and Xception; Ens(02)_m = max of them.

^cEns(12) = Average of InceptionV3, IncepresV2 and Xception; Ens(12)_m = max of them.

^dEns(012) = Average of IncepresV2 and Xception; Ens(012)_m = max of them.

^eEns(3,0) = Average (max) of Ens(012) and InceptionV3.

^fEns(3,1) = Average (max) or max of Ens(012) and IncepresV2.

^gEns(3,2) = Average of Ens(012) and Xception; Ens(3,2)_m = max of them.

^hEns(3,01) = Average of Ens(012) and Ens(01); Ens(3,01)_m = max of them.

ⁱEns(3,02) = Average of Ens(012) and Ens(02); Ens(3,02)_m = max of them.

^jEns(01,02) = Average of Ens(01) and Ens(02); Ens(01,02)_m = max of them.

^kEns(3, Ens(01,02))_m = Average of Ens(012), Ens(01,02)_m; Ens(3, Ens(01,02))_m = max of them.

Deep Learning: Automated identification and grading system of diabetic retinopathy using deep neural networks.

Wei Zhang et. al, Knowledge based systems[6]

Multi Model Feature Fusion Results (Grading system)

Metrics of the grading system ensembles: integration.									
Model	Accuracy	Recall	Precision	Sensitivity	Specificity	Kappa	F1_score	F_{β_score}	Youden's index
Ens(04) ^a	0.9559	0.9292	0.9716	0.9828	0.9885	0.9336	0.9552	0.9506	0.9544
Ens(045) ^b	0.9494	0.9242	0.9448	0.9645	0.9885	0.9240	0.9491	0.9403	0.9530
Ens(0456) ^c	0.9494	0.9242	0.9448	0.9645	0.9885	0.9240	0.9491	0.9403	0.9530
10:Ens(0124567) ^d	0.9637	0.9411	0.9647	0.9810	0.9856	0.9454	0.9632	0.9594	0.9667
Ens(01567) ^e	0.9624	0.9404	0.9634	0.9812	0.9856	0.9435	0.9618	0.9582	0.9667
Ens(01,56) ^f	0.9624	0.9406	0.9632	0.9811	0.9856	0.9435	0.9615	0.9577	0.9667
Ens(8,10) ^g	0.9624	0.9406	0.9632	0.9811	0.9856	0.9435	0.9615	0.9577	0.9667
Ens(Ens(3,01),8,10) ^h	0.9611	0.9353	0.9627	0.9811	0.9856	0.9414	0.9605	0.9564	0.9667
Ens(3,9,567) ⁱ	0.9611	0.9353	0.9627	0.9811	0.9856	0.9414	0.9605	0.9564	0.9667
Ens(Ens(3,01),9,567) ^j	0.9637	0.9412	0.9644	0.9834	0.9856	0.9454	0.9632	0.9564	0.9690
Ens(9,567)	0.9611	0.9428	0.9584	0.9787	0.9856	0.9416	0.9608	0.9550	0.9644

^aEns(04) = Average of InceptionV3 and Resnet50.

^bEns(045) = Average of InceptionV3, Resnet50, DN_169.

^cEns(0456) = Average of InceptionV3, Resnet50, DN_169, and DN_201.

^dEns(0124567) = Average of all models.

^eEns(01567) = Average of InceptionV3, InceptionV3Resnet50, DN_169, and DN_201.

^fEns(01,56) = Average of Ens(01) and Ens(56).

^gEns(8,10) = Average of Ens(456) and Ens(01567).

^hEns(Ens(3,01),8,10) = Average of Ens(3,01), Ens(456) and Ens(01567).

ⁱEns(3,8,567) = Average of Ens(012), Ens(4567) and Ens(567).

^jEns(Ens(3,01),9,567) = Average of Ens(012), Ens(4567) and Ens(567).

Comparison

Journal	Title	Dataset	Preprocessing method	Implemented Method	Performance Metrics	Advantages	Disadvantages
Q1 [3], Retina	Automated diagnosis of retinopathy by content-based image retrieval	- Raw dataset 395 region and lesion based retinal images.	170 independent features representing the number of lesions in a population measured	Content based Image Retrieval (CBIR)	-Sensitivity -Specificity -Accuracy	-Visual Similarity -Reduced Human Error -Less computational Time than deep learning -Indexing using Image data and meta data	- Limited to Known Patterns -Data Quality -Interpretation Challenges -Lack of Clinical Context
Q2 [4], PeerJ Comput Sci.	Detection of diabetic retinopathy using a fusion of textural and ridgelet features of retinal images and sequential minimal optimization classifier.	- DIARETDB1,2007 - KAGGLE, 2019	-Image conversion, - Filtering, -Morphological operations, -Segmentation.	Fusion of textural and ridgelet features and sequential minimal optimization classifier.	-Sensitivity -Specificity -Accuracy -F score	-Doesn't Require large dataset -Avoids underfitting and overfitting problems. -Easily explainable to professional ophthalmologist.	-May include irrelevant features -Incur larger computational time. -Reduced recognition accuracy.
Q1 [5], Artificial Intelligence In Medicine	Referable diabetic retinopathy identification from eye fundus images with weighted path for convolutional neural network.	- Raw Dataset (60,000 images) - STRAE Dataset (131 images)	-Image conversion, Image resizing.	-Weighted path CNN	-Accuracy -AUC -F1 score	-Reduces feature redundancy -Efficient training convergence rate. -Faster training time -Good classification accuracy.	- Huge dataset required for training - Less performance metrics used in the work - Limited to image data
Q1 [6], Knowledge-Based Systems	Automated identification and grading system of diabetic retinopathy using deep neural networks.	- Sichuan Academy of Medical Sciences - Sichuan Provincial Peoples Hospital	- Image Cropping - Image Resizing - Grayscale Conversion - Histogram Equalization.	-Transfer learning models -Ensemble Learning -Customized standard deep neural network	Accuracy,Precision, Sensitivity, Specificity, AUC, Kappa, F1_score, FB_score, Support, Youden's Index	- Two different system can be used for different application requirement -Model Diversity -Improved accuracy -Reduced Risk of False Positives and Negatives	- Blackbox of DNN - Huge computational cost - High computational time - Large dataset required.

- [1] <https://www.magnacode.co.in/blog/myths-and-truths-about-relation-between-diabetes-and-eye-sight>
- [2] Nikos Tsiknakis, Dimitris Theodoropoulos, Georgios Manikis, Emmanouil Ktistakis, Ourania Boutsora, Alexa Berto, Fabio Scarpa, Alberto Scarpa, Dimitrios I. Fotiadis, Kostas Marias, Deep learning for diabetic retinopathy detection and classification based on fundus images: A review, *Computers in Biology and Medicine*, Volume 135, 2021, 104599, ISSN 0010-4825.
- [3] Chaum E, Karnowski TP, Govindasamy VP, Abdelrahman M, Tobin KW. Automated diagnosis of retinopathy by content-based image retrieval. *Retina*. 2008 Nov-Dec;28(10):1463-77. doi: 10.1097/IAE.0b013e31818356dd. PMID: 18997609.
- [4] Ramasamy LK, Padinjappurathu SG, Kadry S, Damaševičius R. Detection of diabetic retinopathy using a fusion of textural and ridgelet features of retinal images and sequential minimal optimization classifier. *PeerJ Comput Sci*. 2021 May 7;7:e456. doi: 10.7717/peerj-cs.456. PMID: 34013026; PMCID: PMC8114804.
- [5] Yi-Peng Liu, Zhanqing Li, Cong Xu, Jing Li, Ronghua Liang, Referable diabetic retinopathy identification from eye fundus images with weighted path for convolutional neural network, *Artificial Intelligence in Medicine*, Volume 99, 2019, 101694, ISSN 0933-3657.
- [6] Wei Zhang, Jie Zhong, Shijun Yang, Zhentao Gao, Junjie Hu, Yuanyuan Chen, Zhang Yi, Automated identification and grading system of diabetic retinopathy using deep neural networks, *Knowledge-Based Systems*, Volume 175, 2019, Pages 12-25, ISSN 0950-7051.

Thankyou

